THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A car covers 90 km in 1.5 h. Its speed is: 1

(a) 135 km/h **(b)** 90 km/h **(c)** 60 km/h **(d)** 45 km/h

2. A car travels 40 km in 1 h and then 60 km in the next hour. Its average speed is: 1

(a) 50 km/h **(b)** 40 km/h **(c)** 60 km/h **(d)** 100 km/h

3. In a distance-time graph, the slope gives: 1

(a) time **(b)** acceleration **(c)** distance only **(d)** speed

4. A runner covers 1.5 km in 10 min. His speed in km/h is: 1

(a) 1.5 km/h **(b)** 0.15 km/h **(c)** 9 km/h **(d)** 15 km/h

5. A bus travels 60 km at 30 km/h. The time taken is: 1

(a) 0.5 h **(b)** 2 h **(c)** 30 h **(d)** 90 h

6. A line in a distance-time graph passing (0,0) and (4 s, 20 m) shows speed: 1

(a) 5 m/s **(b)** 20 m/s **(c)** 4 m/s **(d)** 80 m/s

7. A scooter moves at 36 km/h. Its speed in m/s is: 1

(a) 129.6 m/s **(b)** 3.6 m/s **(c)** 10 m/s **(d)** 36 m/s

8. An athlete runs 100 m in 12.5 s. The speed is: 1

(a) 1250 m/s **(b)** 12.5 m/s **(c)** 8 m/s **(d)** 0.125 m/s

9. Another line passes (0,0) and (2 s, 20 m). Compared with the first, this object is: 1

(a) equal **(b)** slower **(c)** at rest **(d)** faster

10. A speed of 25 m/s equals: 1

(a) 90 km/h **(b)** 25 km/h **(c)** 9 km/h **(d)** 250 km/h

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A car travels 150 km in 2.5 h. Find its speed in km/h and m/s. 2

12. Explain why speed of a cyclist uphill is less than downhill. 2

13. Why should a pendulum be released with a small angle? 2

14. How long will a train at 90 km/h take to cover 225 km? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cyclist covers 12 km in 40 minutes. Find the speed in km/h and m/s. 3

16. A man walks to a market 3 km away at 5 km/h and returns at 3 km/h. Find the total time. 3

17. A student measured the time taken by a ball to roll 2 m as 4 s and then 6 m as 8 s. Compare the speeds. 3

OR

A train 200 m long passes a pole in 20 s. Find its speed in m/s and km/h. 3

18. Why is the speed of a vehicle in cities less than on highways? 3

OR

Why do athletes use electronic timers? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A car starts from city A at 8:00 a.m. and reaches city B 240 km away at 12:00 noon, with a half-hour stop. Find the average speed including the stop and the running speed. 5

OR

Explain why a pendulum clock stops after a while and how it is kept going. 5

20. Draw and explain the distance-time graph of a car moving at 20 km/h for 2 h, resting for 1 h and then moving at 30 km/h for 2 h. 5

OR

Convert units and solve: A cheetah runs 100 m in 4 s; a car goes at 72 km/h. Which is faster? 5

END OF QUESTION PAPER — Total: 40 Marks

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